

Manthan Chawda

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PROFESSIONAL SUMMARY

Computer Science undergraduate with interests in AI/ML, mathematics and software development. Experienced in building practical projects across machine learning, computer vision, software systems, robotics and drones. Interested in applying strong technical foundations to real-world problems while exploring research and new areas of computing.

EDUCATION

GSFC University <i>B.Tech in Computer Science & Engineering — CGPA: 8.47</i>	Vadodara, Gujarat <i>July 2024 – Present</i>
Bright School <i>Higher Secondary & Secondary Education, PCM</i>	Vadodara, Gujarat <i>May 2024</i>

EXPERIENCE

Academic Associate <i>GSFC University</i>	January 2026 – Present <i>Vadodara, Gujarat</i>
- Guide undergraduate students through experimental physics laboratory sessions, explaining concepts and helping bridge the gap between theory and practical application. - Set up and maintain laboratory equipment and supervise hands-on experiments while ensuring proper equipment handling and adherence to laboratory safety procedures.	
Intern <i>GUIITAR Council, GSFC University</i>	December 2025 <i>Vadodara, Gujarat</i>
- Researched aerodynamic and modular drone designs inspired by mini-FPV platforms, with a focus on improving flight efficiency and range. - Analyzed and proposed structural design iterations for lightweight 3D-printed wings and body components to improve aerodynamics and flight duration.	
Secretary, Entrepreneurship Club <i>GSFC University</i>	August 2025 – September 2026 <i>Vadodara, Gujarat</i>
- Coordinate student-led entrepreneurship events, guest sessions and club activities, including planning, documentation and logistics. - Prepare meeting agendas, maintain club records and coordinate communication among students, mentors and club members while supporting event execution.	
Treasurer, Heritage Club <i>GSFC University</i>	August 2025 – August 2026 <i>Vadodara, Gujarat</i>
- Manage club finances, maintain financial records and oversee budgets and expenditures for cultural and heritage activities. - Collaborate with the executive committee to plan budgets and track expenditures for club initiatives and events.	

PROJECTS

CivicPulse <i>YOLO11m-seg, ByteTrack, MiDaS, Python, PyTorch, FastAPI</i>	August 2026 – Present
- Developed an AI-powered civic infrastructure monitoring platform that analyses drone footage to detect potholes, waterlogging, open manholes, drainage overflow and damaged footpaths. - Built a computer vision pipeline using YOLO11m-seg for detection and segmentation, ByteTrack for tracking hazards across video frames and MiDaS for depth estimation. - Designed an incident-based workflow that combines repeated detections into persistent civic incidents and assigns severity and priority using defined scoring logic. - Integrated FastAPI, React/Vite and Supabase/PostgreSQL to process detections, store incident data, associate telemetry and present results through an interactive dashboard.	

ClearContract | *Gemini API, React, Chrome Extension, Node.js, Express, MongoDB* July 2026 – Present

- Developed a Chrome extension prototype that analyses pasted contract and Terms of Service text using Google's Gemini API to identify potentially concerning clauses and explain them in simpler language.
- Built a Node.js and Express backend to handle requests between the browser extension, Gemini API and MongoDB-based application components.
- Implemented the core extension-to-backend analysis workflow and authentication components while continuing to develop the dashboard and broader browser-based analysis features.

Meshmerise | *C++, Arduino IDE, STM32, RLS08, TB6612FNG* October 2025 – December 2025

- Designed and built an autonomous maze-solving robot using an STM32 microcontroller and RLS08 line sensors for navigation.
- Implemented control logic in Arduino IDE for line following and left-hand maze solving across 90-degree mazes.
- Recorded turns during the first traversal and used them to make the second traversal more efficient through path optimization.
- Integrated the TB6612FNG motor driver and regulated power delivery using a buck converter.

ACHIEVEMENTS

First Runner-Up – ELCIA Next-Gen Innovative Tech Hackathon 2026

Smart City Drone-AI Challenge – Team CivicPulse, among 183 teams from 18 states and 100+ institutions.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, Swift, HTML/CSS

AI/ML: PyTorch, TensorFlow, Scikit-learn, OpenCV, NumPy, pandas, Matplotlib

Frameworks: FastAPI

Developer Tools: Git, Docker, VS Code, Xcode, Arduino IDE, Blender, LaTeX

CERTIFICATIONS

Getting Started with Artificial Intelligence – IBM

August 2026

Networking Basics – Cisco Networking Academy

March 2026

TIDES Ignition Program 2026 – IIT Roorkee

January 2026

Oracle Cloud Infrastructure Data Science Professional

October 2025

Oracle Cloud Infrastructure AI Foundations

October 2025

Python Essentials 1 & 2 – Cisco Networking Academy

September 2025

AWS Cloud Foundations – Amazon Web Services

July 2025